STAT ASSESSMENT

Q1. $\bar{X}_1 = 89$ $\quad S_1 = 4$ $\quad n_1 = 50$

$\bar{X}_2 = 82$ $\quad S_2 = 9$ $\quad n_2 = 120$

LOS = 5% = 0.05

$H_0 = \mu_1 = \mu_2$ (No difference)

$H_1 = \mu_1 \neq \mu_2$ (Difference exists)

★ **Z-test Formula:**

- Because sample size is large

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$= \frac{89 - 82}{\sqrt{\frac{(4)^2}{50} + \frac{(9)^2}{120}}}$$

$$= \frac{7}{\sqrt{\frac{16}{50} + \frac{81}{120}}}$$

$$\boxed{Z = 7.02}$$

★ <u>Critical value</u> :-

$LOS = 0.05$

$= \frac{0.05}{2}$ (two tailed)

$= 0.025$

$Z = 1.96$

calculated $Z = 7.02$

Critical $Z = 1.96$

since, $7.02 > 1.96$

$\therefore$ Reject H_0

$\therefore$ There is significance difference between Girls & boys

claim is not valid.

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Q2

Category	cancer	No cancer	Total
Smokers	220	230	550
Non Smokers	350	640	990
Total	680	910	1590

Conclude that smoking causes cancer or not ?

H_0 = Smoking & cancer has no relation

H_1 = Both are related.

★ Expected value (E) = $\frac{\text{Row Total} \times \text{column Total}}{\text{Grand Total}}$

① Smokers with cancer (O = 220)

$$E = \frac{550 \times 680}{1590} = 235.22$$

② ~~Non Smokers with cancer~~

$$\cancel{E = \frac{990 \times 680}{1590} = 444.78}$$

② Smokers without cancer (O = 230)

$$E = \frac{550 \times 910}{1590} = 314.78$$

③ Non smokers with cancer (O = 350)

$$E = \frac{990 \times 680}{1590} = 444.78$$

④ Non smoker without cancer (O = 640)

$$E = \frac{990 \times 910}{1590} = 565.22$$

★ Chisqware Formula :-

$$X^2 = \sum \frac{(O-E)^2}{E}$$

① Smoker + cancer = 0.98
② Smoker + Non cancer = 22.8
③ Non smoker + cancer = 20.2
④ Non smoker + No cancer = 9.9

$$X^2 = 0.98 + 22.8 + 20.2 + 9.9$$
$$= 63.9$$

☆ Critical value

Degree of freedom =

$df = (2-1)(2-1)$
$= 1$

At 5% = 3.84

Calculated = 63.9
Critical = 3.84

$63.9 > 3.84$ → Reject H_0

→ Smoking & cancer are related

∴ Smoking does cause cancer.